

Chapter 10 - Libraries

A library is code that everyone can use to add certain functionality to their project.

Let's try out a library. We're going to use *tick* by *rxl*. You can find the library on [GitHub](#).

Click on tick.lua and then on Raw, and [copy the code](#).

Go to your text editor, create a new file called tick.lua and paste the code.

Now we have to follow the instructions on the GitHub page. First we have to load it with require.

```
function love.load()
```

```
    tick = require "tick"
```

```
end
```

Notice how require doesn't have parantheses (). This is because when you only pass 1 argument, you don't have to use them. Now I recommend you still do use them for any other function, but with require it's common to leave them out. But in the end, it doesn't even matter.

Next we have to put tick.update(dt) in our updater.

```
function love.update(dt)
```

```
    tick.update(dt)
```

```
end
```

And now we're ready to start using the library. Let's make it so that a rectangle will be drawn after 2 seconds.

```
function love.load()
```

```
    tick = require "tick"
```

```
    --Create a boolean
```

```
    drawRectangle = false
```

```
    --The first argument is a function
```

--The second argument is the time it takes to call the function

```
tick.delay(function () drawRectangle = true end , 2)
```

```
end
```

```
function love.draw()
```

--if drawRectangle is true then draw a rectangle

```
if drawRectangle then
```

```
    love.graphics.rectangle("fill", 100, 100, 300, 200)
```

```
end
```

```
end
```

Did we just pass a function as an argument? Sure, why not? A function is a type of variable after all.

When you run the game you can see that with this library we can put a delay on calling functions. And like that there are tons of libraries with all kinds of functionality.

Don't feel guilty for using a library. Why reinvent the wheel? That is, unless you are interested in learning it. I personally use about 10 libraries in my projects. They provide functionality that I don't understand how to make myself, and I'm simply not interested in learning it.

Libraries aren't magic. It's all Lua code that you and I could've written as well (in case we have the knowledge of course). We'll create a library in a future chapter so that we have a better understanding of how they work.

Standard libraries

Lua has built-in libraries. These are called *Standard Libraries*. They are the functions that are built into Lua. `print`, for example, is part of the standard library. And so is `table.insert` and `table.remove`.

One important standard library we haven't looked at yet is the *math library*. It provides math functions, which can be very useful when making a game.

For example, `math.random` gives us a random number. Let's use it to place a rectangle on a random position whenever you press the spacebar.

```
function love.load()
```

```
    x = 30
```

```
    y = 50
```

```
end
```

```
function love.draw()
```

```
    love.graphics.rectangle("line", x, y, 100, 100)
```

```
end
```

```
function love.keypressed(key)
```

```
    --If space is pressed then..
```

```
    if key == "space" then
```

```
        --x and y become a random number between 100 and 500
```

```
        x = math.random(100, 500)
```

```
        y = math.random(100, 500)
```

```
    end
```

```
end
```

Now that we understand what libraries are, we can start using a class library.

Summary

Libraries are code that gives us functionality. Anyone can make a library. Lua also has built-in libraries which we call the Standard Libraries.